UNIT IV : Quadrilateral Elements & Thermal Analysis

4.1 Isoparametric, Sub-parametric and Super-parametric elements

A general quadrilateral with straight edges cannot be described by a simple polynomial in x, y. The trick is to work in a **parent (natural) element** — a square $-1 \le \xi, \eta \le 1$ — and **map** it to the real element. Two sets of interpolation are used :

$$x = \Sigma N_i^g(\xi,\eta)\, x_i,\ y = \Sigma N_i^g\, y_i \text{ (geometry)}\quad u = \Sigma N_i^u(\xi,\eta)\, u_i \text{ (displacement)}$$

Type	Geometry nodes vs displacement nodes	Result
Isoparametric	same shape functions for both ($N^g = N^u$)	most widely used
Sub-parametric	geometry order < displacement order	curved geometry not needed, accurate u
Super-parametric	geometry order > displacement order	rarely used, may fail patch test

"Iso" = same, "sub" = fewer nodes for geometry than for the field, "super" = more nodes for geometry than for the field. Example : a 8-noded element with a straight-edge geometry described by 4 corner nodes only = **sub-parametric**; a 4-noded displacement field with 8-noded curved geometry = **super-parametric**.

Why use them?: Isoparametric elements pass the patch test and satisfy completeness / compatibility if the mapping is one-to-one (det $J > 0$ everywhere, element convex, interior angles < 180°).

4.2 Four-noded Quadrilateral (Q4)

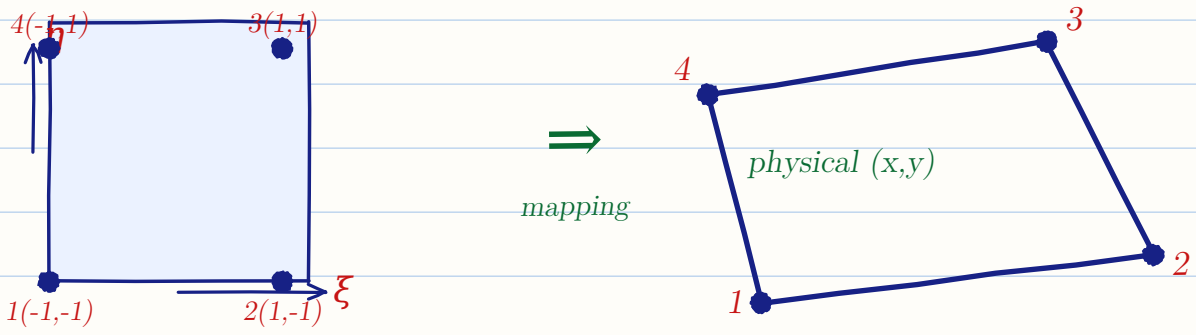


Fig 4.1 Q4 : parent element (ξ,η) and physical element (x,y)

Shape functions (bilinear; $N_i = 1$ at node i, 0 at other nodes) with node numbering counter-clockwise 1(−1,−1), 2(1,−1), 3(1,1), 4(−1,1) :

$$N_1 = (1-\xi)(1-\eta)/4\quad N_2 = (1+\xi)(1-\eta)/4\quad N_3 = (1+\xi)(1+\eta)/4\quad N_4 = (1-\xi)(1+\eta)/4$$

Properties : $\Sigma N_i = 1$; each N_i is linear along every edge ⇒ displacement compatible between neighbours. Displacement field has 8 dof : $\{q\} = [u_1\ v_1\ \ldots\ u_4\ v_4]^T$, contains terms 1, ξ, η, $\xi\eta$.

$$\{u,v\}^T = \begin{bmatrix} N_1 & 0 & N_2 & 0 & N_3 & 0 & N_4 & 0 \\ 0 & N_1 & 0 & N_2 & 0 & N_3 & 0 & N_4 \end{bmatrix}$$

Jacobian, [B] and [k]

Derivatives of N_i are known in (ξ,η) but strains need (x,y). Chain rule :

$$\{\partial/\partial\xi\ ;\ \partial/\partial\eta\} = [J]\{\partial/\partial x\ ;\ \partial/\partial y\}, \quad [J] = \begin{bmatrix} \partial x/\partial\xi & \partial y/\partial\xi \\ \partial x/\partial\eta & \partial y/\partial\eta \end{bmatrix}$$

$$[J] = [dN/d\xi\ ;\ dN/d\eta]\ [x_1\ y_1\ ;\ \ldots\ ;\ x_4\ y_4]\ \{\partial N/\partial x\ ;\ \partial N/\partial y\} = [J]^{-1}\ \{\partial N/\partial\xi\ ;\ \partial N/\partial\eta\}$$

$$[B] = \begin{bmatrix} N_{1,x} & 0 & N_{2,x} & 0 & N_{3,x} & 0 & N_{4,x} & 0 \\ 0 & N_{1,y} & 0 & N_{2,y} & 0 & N_{3,y} & 0 & N_{4,y} \\ N_{1,y} & N_{1,x} & N_{2,y} & N_{2,x} & N_{3,y} & N_{3,x} & N_{4,y} & N_{4,x} \end{bmatrix}$$

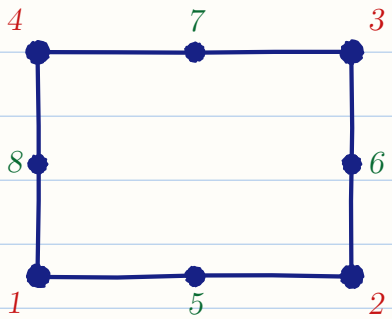
$$[k] = t \int_{-1}^{1}\int_{-1}^{1} [B]^T[D][B]\ det[J]\ d\xi\ d\eta$$

$dx\ dy = det[J]\ d\xi\ d\eta$. The integrand is a rational function of ξ, η (because of J^{-1}) for a general quad, so the integral is evaluated **numerically** (Gauss quadrature). For a rectangle J is constant and the integrand is polynomial.

Load vectors

- Body force : $\{f_b\} = t \iint [N]^T\{X\ Y\}^T\ det\ J\ d\xi\ d\eta$ (for uniform force on a rectangle, one quarter of the total at each node).
- Edge traction on side with nodes i-j : $\{f_T\} = t \int [N]^T\{T_x\ T_y\}^T\ ds$, $ds = (L_{ij}/2)\ d\xi$ ⇒ uniform traction gives $T\ t\ L_{ij}/2$ at each of the two nodes.

4.3 Eight-noded Quadrilateral (Q8, serendipity)



corner nodes 1-4

mid-side nodes 5-8

16 dof, curved edges

Fig 4.2 Q8 : corner nodes 1-4, mid-side nodes 5-8

Q8 has curved edges (parabolic) and 16 dof; polynomial contains 1, ξ, η, ξ^2, $\xi\eta$, η^2, $\xi^2\eta$, $\xi\eta^2$. Shape functions (serendipity family) :

Corner nodes *($i = 1..4$, ξ_i, $\eta_i = \pm 1$) : $N_i = (1/4)(1 + \xi\xi_i)(1 + \eta\eta_i)(\xi\xi_i + \eta\eta_i - 1)$*

Mid-side nodes *: $N_5 = \frac{1}{2}(1-\xi^2)(1-\eta)$, $N_6 = \frac{1}{2}(1+\xi)(1-\eta^2)$, $N_7 = \frac{1}{2}(1-\xi^2)(1+\eta)$, $N_8 = \frac{1}{2}(1-\xi)(1-\eta^2)$.*

Check : at node 5 (0,−1) $N_5 = \frac{1}{2}(1)(2) = 1$; sum of all $N = 1$; each N_i varies quadratically on edges, so the strain varies linearly - much better than Q4 for bending and curved boundaries. Stiffness : same formulae with 16×16 [k], integrated with 3×3 Gauss points.

> ***Remark:*** *Q4 shows shear locking in pure bending (too stiff), Q8 is far better. Q9 (Lagrange, adds centre node) has shape functions N = products of 1-D quadratic Lagrange polynomials.*

4.4 Numerical Integration (Gauss Quadrature)

Gauss-Legendre rule : $\int_{-1}^{1} f(\xi)\, d\xi \approx \Sigma\, w_i\, f(\xi_i)$. With n points it integrates exactly a polynomial of degree ***2n − 1****.*

n	Gauss points ξ_i	Weights w_i	Exact up to degree
1	0	2	1
2	±0.57735	1 , 1	3
3	0 , ±0.77460	8/9 , 5/9	5
4	±0.33998 , ±0.86114	0.65215 , 0.34785	7

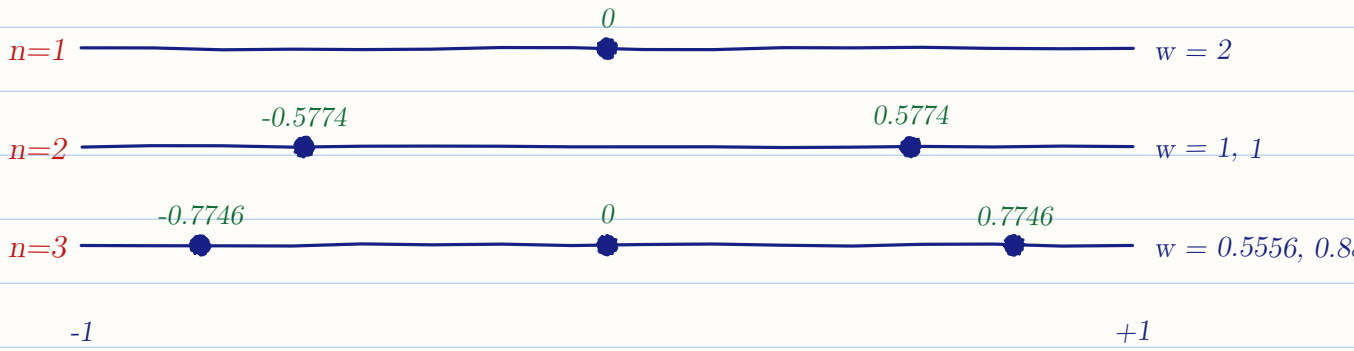


Fig 4.3 Gauss points on [−1, 1]

Two dimensions *: $\int\int f\, d\xi\, d\eta \approx \Sigma\Sigma\, w_i w_j\, f(\xi_i, \eta_j)$. For Q4 use 2×2 (4 points, w*

= 1, points (±0.57735, ±0.57735)); for Q8 use 3×3 (9 points).

General interval [a,b] : $x = (b-a)\xi/2 + (a+b)/2$, $dx = (b-a)/2\, d\xi$.

Example : Gauss integration

Evaluate $I = \int_{-1}^{1} (x^3 + 2x^2 + 1)\, dx$. Exact = 0 + 4/3 + 2 = 3.3333. Two-point rule : f(−0.57735) + f(0.57735) = 1.4742 + 1.8591 = 3.3333 (exact, since degree 3 ≤ 2n − 1 = 3).

One-point rule would give 2 f(0) = 2 (wrong, error 40 %). For $\int_1^3 x^2\, dx$: $x = \xi + 2$, $dx = d\xi \Rightarrow \int (\xi+2)^2 d\xi = 2.0 \cdot [f(-0.577)+f(0.577)]$ with $f = (\xi+2)^2$: 2.0204 + 6.6463 = 8.6667 = 26/3 (exact).

4.5 Solved Problems on the Quadrilateral

Problem 1 : mapping and Jacobian

Quadrilateral nodes 1 (0,0), 2 (4,0), 3 (5,3), 4 (1,3). (a) Find the global coordinates of the point $(\xi,\eta) = (0.5, 0.5)$; (b) Jacobian at that point.

$N = [0.0625, 0.1875, 0.5625, 0.1875]$: $N_1 = (0.5)(0.5)/4 = 0.0625$, $N_2 = (1.5)(0.5)/4 = 0.1875$, $N_3 = (1.5)(1.5)/4 = 0.5625$, $N_4 = (0.5)(1.5)/4 = 0.1875$.

$x = \Sigma N_i x_i = 0.1875\times4 + 0.5625\times5 + 0.1875\times1 = 3.7500$, $y = 0.5625\times3 + 0.1875\times3 = 2.2500$.

$$[J] = \begin{bmatrix} 2.000 & 0.000 \\ 0.500 & 1.500 \end{bmatrix}$$

$\det[J] = 2.000 \times 1.500 - 0.000 \times 0.500 = 3.000$ (> 0 ⇒ valid mapping). Check : det J > 0 at all four Gauss points as well; the element is convex.

Area of the quadrilateral = $\int\int \det J\, d\xi\, d\eta$ = 2×2 Gauss sum = 12.000 mm^2 (trapezium : ½(4+4)×3 = 12 OK.)

Problem 2 : stiffness matrix of a square Q4 element

Square of side a (any size), plane stress, thickness t, $\nu = 0.3$, 2×2 Gauss integration. det $J = a^2/4$ is constant, $J = (a/2)\, I$. The result is $k = E\, t \times$ (matrix below) :

$$[k]/(Et) = \begin{bmatrix} 0.495 & 0.179 & -0.302 & -0.014 & -0.247 & -0.179 & 0.055 & 0.014 \\ 0.179 & 0.495 & 0.014 & 0.055 & -0.179 & -0.247 & -0.014 & -0.302 \\ -0.302 & 0.014 & 0.495 & -0.179 & 0.055 & -0.014 & -0.247 & 0.179 \\ -0.014 & 0.055 & -0.179 & 0.495 & 0.014 & -0.302 & 0.179 & -0.247 \\ -0.247 & -0.179 & 0.055 & 0.014 & 0.495 & 0.179 & -0.302 & -0.014 \\ -0.179 & -0.247 & -0.014 & -0.302 & 0.179 & 0.495 & 0.014 & 0.055 \\ 0.055 & -0.014 & -0.247 & 0.179 & -0.302 & 0.014 & 0.495 & -0.179 \\ 0.014 & -0.302 & 0.179 & -0.247 & -0.014 & 0.055 & -0.179 & 0.495 \end{bmatrix}$$

Properties : symmetric ; each row sums to zero for the x-dof rows (rigid translation) ; diagonal terms positive ($k_{11} = 0.4945\ Et$). The matrix is independent of the size a for a square (stiffness $\propto t$) - a characteristic of 2-D elasticity.

4.6 Steady-state Heat Transfer

*Heat is transferred by **conduction** (Fourier : $q = -k\ dT/dx$), **convection** (Newton : $q = h(T_s - T_\infty)$) and radiation. In FEM the field variable is the temperature T, one dof per node. Governing equation (1-D, steady, with heat generation Q) :*

$$d/dx\ (\ k\ A\ dT/dx\ ) + Q = 0 \quad BC : T = T_0 \text{ (prescribed) or } -k\ dT/dx = h\ (T - T_\infty) \text{ (convection) or } q = q_0$$

Piece-wise linear temperature in a 2-noded element : $T = N_1T_1 + N_2T_2$, $N_1 = (1-\xi)/2$, $N_2 = (1+\xi)/2$. Applying the functional or Galerkin method :

$$[K_c] = kA/L \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} T_1 \\ T_2 \end{Bmatrix} = (QAL/2)[1,1]^T + \{q\}$$

Analogy with the bar : displacement ↔ temperature, force ↔ heat flow, AE/L ↔ kA/L. Hence the assembly and BC procedure are the same as Unit I.

- ***Convection at a node** (surface area A) : add hA to the diagonal term K_{ii} and $hA\ T_\infty$ to the load F_i.*
- ***Prescribed heat flux** q_0 at a node : add q_0A to F_i. **Insulated end** : nothing to add (natural BC).*
- *Heat flow through an element $Q = -kA\ (T_2 - T_1)/L$.*

Composite slab (wall)

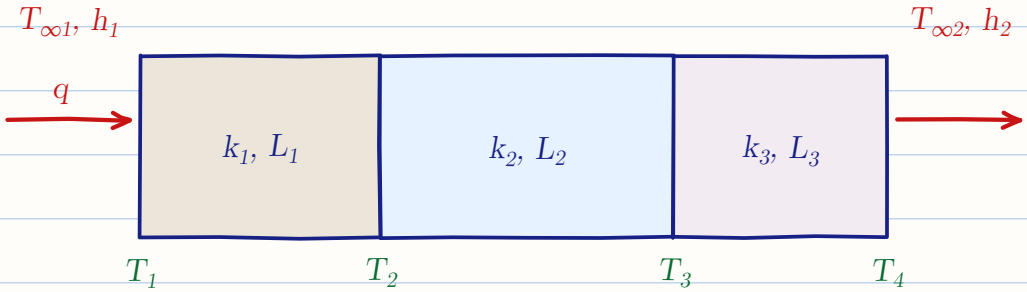


Fig 4.4 Three-layer composite wall

Each layer = one element of conductance k_iA/L_i; nodes at the interfaces. Both outer surfaces have convection (h_1, $T_{\infty 1}$ and h_2, $T_{\infty 2}$). Global equation for 3 layers (4 nodes) :

$$[K]\ (A=1)\ ;\ k_i = k_i/L_i \begin{bmatrix} k_1+h_1 & -k_1 & 0 & 0 \\ -k_1 & k_1+k_2 & -k_2 & 0 \\ 0 & -k_2 & k_2+k_3 & -k_3 \\ 0 & 0 & -k_3 & k_3+h_2 \end{bmatrix}$$

with load vector $F = [h_1T_{\infty 1},\ 0,\ 0,\ h_2T_{\infty 2}]^T$ (area $A = 1\ m^2$).

Problem 3 : numerical (3-layer wall)

Wall of layers $k = 20, 30, 50\ W/m \cdot K$, thickness 0.30, 0.15, 0.15 m. Inside gas $T_{\infty 1} = 800\ °C$, $h_1 = 25\ W/m^2K$; outside air $T_{\infty 2} = 20\ °C$, $h_2 = 15\ W/m^2K$. Area $= 1\ m^2$. Find interface temperatures and heat flux.

$k_1/L_1 = 20/0.3 = 66.67$, $k_2/L_2 = 30/0.15 = 200.00$, $k_3/L_3 = 50/0.15 = 333.33$ W/K.

$$\begin{bmatrix} 91.67 & -66.67 & 0.00 & 0.00 \\ -66.67 & 266.67 & -200.00 & 0.00 \\ 0.00 & -200.00 & 533.33 & -333.33 \\ 0.00 & 0.00 & -333.33 & 348.33 \end{bmatrix} \begin{Bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{Bmatrix} = [20000,\ 0,\ 0,\ 300]^T$$

Solving : $T_1 = 559.38\ °C$, $T_2 = 469.15\ °C$, $T_3 = 439.07\ °C$, $T_4 = 421.03\ °C$. Heat flux $q = h_1(T_{\infty 1} - T_1) = 6015.4\ W/m^2$. Check with the resistance method : $q = (800 - 20)/(1/25 + 0.3/20 + 0.15/30 + 0.15/50 + 1/15) = 6015.4\ W/m^2$ OK.

4.7 Fin (Extended Surface) Analysis

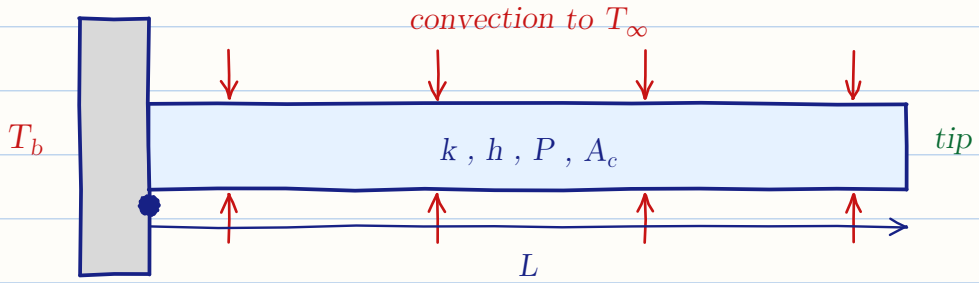


Fig 4.5 Straight fin attached to a hot wall

Fin governing equation (1-D, constant section, perimeter P, area A_c, convection h to ambient T_∞) :

$$k\,A_c\,d^2T/dx^2 - h\,P\,(T - T_\infty) = 0 \quad T(0) = T_b \text{ (base); tip : insulated or } -k\,dT/dx = h(T - T_\infty)$$

Element (2-noded) = conduction part + convection part :

$$[k_e] = kA_c/L \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} + hPL/6 \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$\{f_e\} = hPT_\infty L/2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Tip convection (no insulation) : add hA_c to the last diagonal term and hA_cT_∞ to the last load term. Heat dissipated by the fin = heat conducted at the base : $Q = -kA_c(dT/dx)_{x=0}$; best evaluated from the ***reaction*** *at node 1 ($R_1 = [K]_{1,*}\{T\} - F_1$) because the element gradient is less accurate than the nodal values.*

Problem 4 : pin fin (insulated tip)

Aluminium-type pin fin : d = 10 mm, L = 100 mm, k = 60 W/m · K, h = 25 W/m^2K, T_b = 120 °C, T_∞ = 30 °C. In mm-W-K units : k = 0.06 W/mm · K, h = 25×10^{-6} W/mm^2K, $P = \pi d$ = 31.42 mm, A_c = 78.54 mm^2. Four elements, L_e = 25 mm.

kA_c/L_e = 0.1885 ; $hPL_e/6$ = 0.00327 ; $hPT_\infty L_e/2$ = 0.29452 W. The element matrix is $[k_e]$ = [0.1950, -0.1852 ; -0.1852, 0.1950] and the assembled system with T_1 = 120 °C gives :

x (mm)	0	25	50	75	100
T FEM (°C)	120.00	99.21	85.76	78.22	75.80
T exact (°C)	120.00	99.33	85.94	78.44	76.02

The FEM solution agrees with the analytical result $T = T_\infty + (T_b - T_\infty)\cosh m(L-x)/\cosh mL$, where $m = \sqrt{(hP/kA_c)} = 0.01291\ mm^{-1}$. Base heat flow from the reaction at node 1, $R = [K]_{1,*}\{T\} - F_1$: 4.734 W (exact 4.705 W).

Fin parameters: Efficiency of the fin $\eta = Q_{actual} / (h\ P\ L\ (T_b - T_\infty))$ and effectiveness $\varepsilon = Q_{fin}/(h\ A_c(T_b - T_\infty))$. Fins are useful when hP/kA_c is small, i.e. high conductivity and slender.

4.8 More Solved Problems

Problem 5 : strain and stress in a Q4 element

Rectangle 2 × 1 mm : nodes (0,0), (2,0), (2,1), (0,1); plane stress $E = 2\times10^5$, $\nu = 0.3$. Nodes 1, 4 fixed; $u_2 = u_3 = 0.002$ mm, $v_3 = -0.0006$ mm, $v_2 = 0$. At the centre ($\xi = \eta = 0$) :

$[J] = [1\ 0\ ;\ 0\ 0.5]$, $\det J = 0.5$, $J^{-1} = [1\ 0\ ;\ 0\ 2]$. $\partial N_i/\partial\xi = [-1, 1, 1, -1]/4$, $\partial N_i/\partial\eta = [-1, -1, 1, 1]/4 \Rightarrow \partial N_i/\partial x = [-0.25, 0.25, 0.25, -0.25]$, $\partial N_i/\partial y = [-0.5, -0.5, 0.5, 0.5]$.

Strains : $\varepsilon_x = \Sigma\ N_{i,x}u_i = 0.25\times0.002 + 0.25\times0.002 = 0.001000$, $\varepsilon_y = \Sigma\ N_{i,y}v_i = 0.5\times(-0.0006) = -0.000300$, $\gamma_{xy} = \Sigma(N_{i,y}u_i + N_{i,x}v_i) = -0.000150$.

Stress $\{\sigma\} = [D]\{\varepsilon\}$: $\sigma_x = 200.00$, $\sigma_y = 0.00$, $\tau_{xy} = -11.54\ N/mm^2$.

Problem 6 : slab with heat generation

Plate $L = 0.1$ m, $k = 20\ W/m\cdot K$, generation $Q = 2\times10^5\ W/m^3$, both faces held at 100 °C. Two elements, $L_e = 0.05$ m, $A = 1\ m^2$. Conduction $k/L_e = 400\ W/K$.

$$[K]\begin{bmatrix} 400 & -400 & 0 \\ -400 & 800 & -400 \\ 0 & -400 & 400 \end{bmatrix}\begin{Bmatrix} T_1 \\ T_2 \\ T_3 \end{Bmatrix} = [5000, 10000, 5000]^T$$

($F_i = Q\ A\ L_e/2$ per element end). With $T_1 = T_3 = 100$: $800\ T_2 = 10000 + 400(100) + 400(100) \Rightarrow T_2 = 112.5$ °C. Exact maximum $T = T_w + QL^2/(8k) = 100 + 2\times10^5\times0.01/160 = 112.5$ °C (exact at the node).

Problem 7 : same fin with convection at the tip

Add $h\,A_c = 25\times10^{-6}\times78.54 = 1.96\times10^{-3}$ W/K to K_{55} and $hA_cT_\infty = 0.0589$ W to F_5. Solution (°C) :

x (mm)	0	25	50	75	100
T (tip convecting)	120.00	98.97	85.26	77.40	74.56
T (insulated tip)	120.00	99.21	85.76	78.22	75.80

Tip temperature is slightly lower with a convecting tip (extra heat loss), but the difference is small for a slender fin; the exact corrected-length method ($L_c = L + d/4$) predicts the same trend.

Two-dimensional heat conduction (triangle) - outline

Temperature $T = N_1T_1 + N_2T_2 + N_3T_3$ (CST shape functions). Gradient $\{g\} = [\partial T/\partial x,\ \partial T/\partial y]^T = [B]\{T\}$, $B = (1/2A)[b_1\ b_2\ b_3\ ;\ c_1\ c_2\ c_3]$.

$$[K_c] = \int [B]^T\, k\, [B]\, t\, dA = (k\, t/4A)\ (\ \{b\}\{b\}^T + \{c\}\{c\}^T\ )$$

Example : triangle (0,0), (2,0), (0,2), $k = 1$, $t = 1$, $A = 2 \Rightarrow b = [-2,\ 2,\ 0]$, $c = [-2,\ 0,\ 2]$:

$$[K_c] = \begin{bmatrix} 1.00 & -0.50 & -0.50 \\ -0.50 & 0.50 & 0.00 \\ -0.50 & 0.00 & 0.50 \end{bmatrix}$$

Convection on an edge i-j : $[K_h] = (h\ t\ L_{ij}/6)\ [2\ 1\ ;\ 1\ 2]$ and $\{f_h\} = (h\ t\ L_{ij}\ T_\infty/2)\ [1\ 1]^T$. Heat generation : $f = Q\ t\ A/3$ each node. The assembly and BC procedure is identical to the structural case.

Structural ↔ thermal analogy

Structural (bar)	Thermal (1-D conduction)
displacement u	temperature T
force F	heat flow Q
stiffness AE/L	conductance kA/L
spring support	convection hA
body force	heat generation Q

Further remarks on quadrilaterals

- **Reduced integration** (1×1 for Q4) is cheaper but gives zero-energy **hourglass modes**; 2×2 is the standard full integration. Q8 with 2×2 (reduced) is sometimes used to avoid locking.
- A quad can be split into 2 triangles but one gets the CST accuracy; a Q4 represents linear strain variation in part (ε_x varies with y).
- Mesh distortion degrades accuracy; keep interior angles between 45° and 135°, aspect ratio < 5, and avoid concave quadrilaterals (det J changes sign).

Gauss quadrature in 2-D : worked example

Evaluate $I = \int_{-1}^{1}\int_{-1}^{1} (\xi^2\eta^2 + \xi\eta + 1)\, d\xi\, d\eta$. Exact : (2/3)(2/3) + 0 + 4 = 4.4444. 2×2 Gauss (all weights 1, points ±0.57735) :

Point	ξ	η	$f(\xi,\eta)$
1	−0.57735	−0.57735	1.4444
2	0.57735	−0.57735	0.7778
3	0.57735	0.57735	1.4444
4	−0.57735	0.57735	0.7778

Sum = 4.4444 ⇒ I = 4.4444 (exact, since the integrand has degree ≤ 3 in each variable).

Equivalent nodal loads on a Q4 edge

Uniform pressure p on side 2-3 ($\xi = +1$) of length L_{23} = 3 mm, t = 2 mm, normal in +x direction : p = 10 N/mm^2. Along the edge $N_2 = (1-\eta)/2$, $N_3 = (1+\eta)/2$, $ds = (L_{23}/2)d\eta$:

$$f_{2x} = f_{3x} = p\, t\, L_{23}/2 = 10 \times 2 \times 3/2 = 30 \text{ N (y-forces zero)}$$

For a **linearly varying** edge load with $p_2 \to p_3$: $f_2 = tL(2p_2 + p_3)/6$, $f_3 = tL(p_2 + 2p_3)/6$. For the 8-noded element a uniform edge load is shared in the ratio 1 : 4 : 1 (×pLt/6) between corner, mid-side and corner nodes - **not** equally.

Q8 : numerical check of the shape functions

At $(\xi,\eta) = (0.5, 0.5)$: $N_1 = -0.1250$, $N_2 = -0.1875$, $N_3 = 0.0000$, $N_4 = -0.1875$, $N_5 = 0.1875$, $N_6 = 0.5625$, $N_7 = 0.5625$, $N_8 = 0.1875$. Sum = 1.0000 (partition of unity OK). Corner shape functions can be negative inside the element - normal for quadratic elements.

Thermal resistance relations (useful check for FEM results)

Mode	Resistance (per unit area A)	FEM equivalent
Conduction (plane wall)	$R = L/(kA)$	$K = kA/L$
Convection	$R = 1/(hA)$	$K = hA$
Series layers	$R_{tot} = \Sigma R_i$	assembled [K]
Heat flow	$Q = \Delta T/R_{tot}$	$Q = K\,\Delta T$ (series)
Overall U	$1/UA = R_{tot}$	—

Analytical fin results (for comparison with FEM)

Tip condition	Temperature $\theta = T - T_\infty$	Heat loss Q
Insulated tip	$\theta_b \cosh m(L-x)/\cosh mL$	$\sqrt{(hPkA_c)}\,\theta_b \tanh mL$
Convecting tip	$\theta_b [\cosh m(L-x) + (h/mk) \sinh m(L-x)] / [\cosh mL + (h/mk) \sinh mL]$	$\sqrt{(hPkA_c)}\,\theta_b [\sinh mL + (h/mk) \cosh mL]/[\cosh mL + (h/mk) \sinh mL]$
Very long fin	$\theta_b\, e^{-mx}$	$\sqrt{(hPkA_c)}\,\theta_b$

with $m = \sqrt{(hP/kA_c)}$. Efficiency of an insulated-tip fin $\eta = \tanh(mL)/(mL)$.

Example above : $mL = 1.291 \Rightarrow \eta = 0.859/1.291 = 0.665$.

Units: Units : keep one consistent system. With mm, N, W : k in $W/mm \cdot K = (W/m \cdot K)/1000$, h in $W/mm^2K = (W/m^2K)/10^6$. A frequent error is mixing k in $W/m \cdot K$ with lengths in mm.

Short answer questions

Q1 Number of Gauss points for a 4-noded quad ? - $2 \times 2 = 4$. For 8-noded ? - $3 \times 3 = 9$.

Q2 What is the Jacobian ? - The matrix relating derivatives in natural and global coordinates; det J = area scale factor.

Q3 Write N_3 of Q4. - $(1+\xi)(1+\eta)/4$.

Q4 What is the order of exactness of 2-point Gauss rule ? - Polynomials up to degree 3.

Q5. What is heat transfer coefficient h ? - Rate of convective heat transfer per unit area per kelvin, W/m^2K.

Expected questions

- Define isoparametric, sub-parametric, super-parametric elements. Derive shape functions of Q4 / Q8; evaluate Jacobian; derive [k] with Gauss integration.
- Gauss quadrature : points and weights for 2 and 3 points; integrate given function; why 2×2 for Q4 ?
- Derive the 1-D conduction element; solve composite wall; derive element matrices of a fin and find temperature distribution.